

Citation Previews in a Research Reader

Fay Instance
School of Worked Examples

Abstract

Following a citation or a figure reference in a PDF usually means leaving the current page. We describe a reader that shows the target in a small preview instead, and compare two ways of finding the target: following the link annotations that the typesetting system wrote into the file, and searching the text for patterns such as bracketed numbers when no links exist.

1 Introduction

Reference previews are an old idea in hypertext systems [1] and have been studied for web pages and wikis [2, 3]. For scholarly PDFs the situation is less settled, because many files carry no link annotations at all and the reader has to recover the structure from the text alone.

The model we use is summarised in Section 2, and the preview pipeline is shown in Figure 1 on the next page.

2 Model

We score a candidate target t for a reference r by combining a textual match with a penalty for distance in the document:

$$s(r, t) = m(r, t) - \lambda d(r, t). \quad (1)$$

The textual match is one when the label of the target equals the visible text of the reference and zero otherwise. The distance is counted in pages, and the weight is chosen so that an exact match always wins:

$$0 < \lambda < \frac{1}{N}. \quad (2)$$



Figure 1: The preview pipeline: detect the reference, locate its target, and render the target region in a popup.

3 Experiments

We ran the scoring rule of Eq. (1) on a set of papers with and without link annotations. Link annotations were present in most recent papers, in line with earlier surveys [4], but almost absent from scanned and older material [5, 6]. The resulting accuracy is shown in Figure 2.

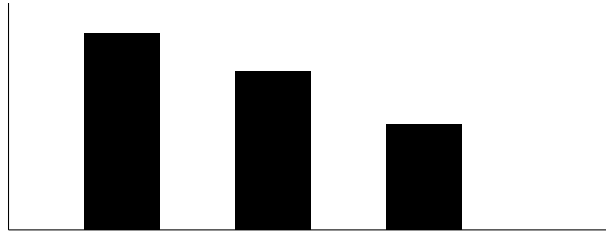


Figure 2: Accuracy of target detection for linked papers, unlinked papers and scanned papers after text recognition.

4 Discussion

The bound in Eq. (2) matters most for unlinked papers, where the same label can occur more than once. As argued in Section 1, links should be preferred whenever they exist; the text patterns of Fig. 1 are a fallback. Similar fallbacks are used by reference managers [7] and by accessibility tools [8].

References

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